## Why not use a computer? It's much more powerful

Even though a computer is powerful, a microcontroller is much more preferable. Why?

- Microcontroller-based devices are very small in size.
- Implementation costs less.
- It's simple to implement and, if necessary, can communicate with a computer.
- A computer will need to be switched on all the time, guzzling power for the processor, hard drive etc.

Microcontroller made the gaming console a reality

- A crash can spell disaster for critical applications. Since microcontrollers run only one application, they may never crash. Problems only occur when they're used in adverse conditions or due to errors on the part of the programmer.
- Your computer has many applications running simultaneously. In critical operations where continuous monitoring is necessary, missing an event because of simultaneously running multiple applications can be undesirable and sometimes catastrophic.
- By interfacing a high speed microprocessor to slow devices the processor will be wasting precious processing power just waiting for data from the sensor.

To sum up the chapter you just read, we can say that microcontrollers are small and powerful chips that supplement the use of a computer to monitor the environment around us and control some aspects of the physical world. They've been used by people the world over to develop many interesting applications.